

- 2 A helium atom may be modelled as a nucleus surrounded by two electrons in diametrically opposite circular orbits, each of radius 170 pm, as shown in Fig. 2.1.

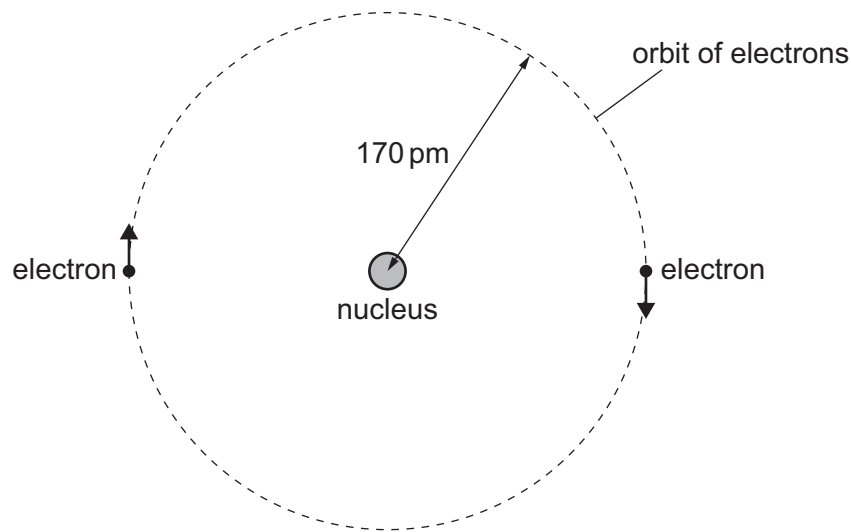


Fig. 2.1

- (a) State Coulomb's law.

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 [2]

- (b) (i) State the charge on the nucleus, in terms of the elementary charge e .

charge = e [1]

- (ii) Show that the electric force between the nucleus and one of the electrons is $1.6 \times 10^{-8} \text{ N}$.

[1]

